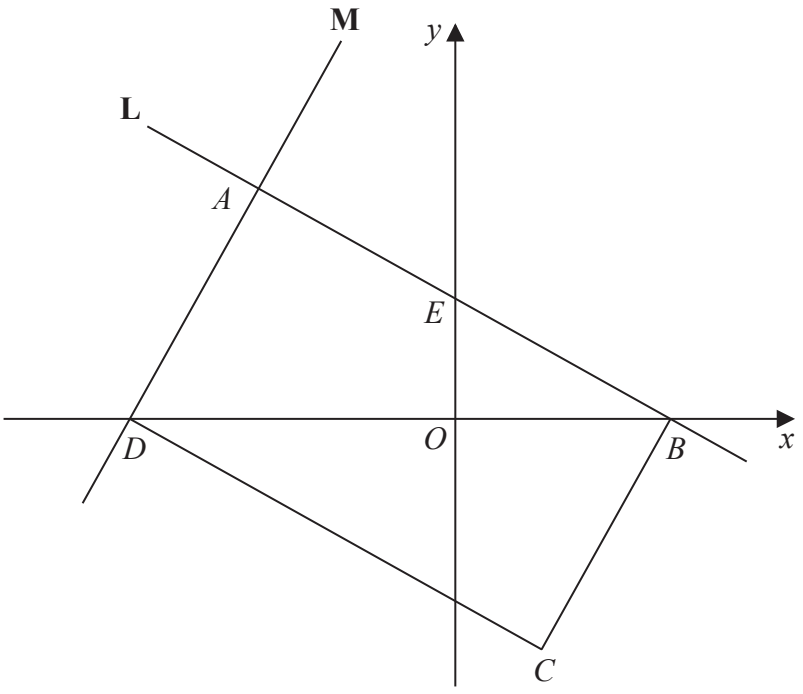
19



ABCD is a rectangle.

A, *E* and *B* are points on the straight line **L** with equation $x + 2y = 12$
A and *D* are points on the straight line **M**.

$AE = EB$

Find an equation for **M**.

L: y = 6 – ½x, so its gradient is –½.

1 mark
Find the gradient of L

B = (12, 0) and E = (0, 6). E is the midpoint of AB.

A = (2 × 0 – 12, 2 × 6 – 0) = (–12, 12).

Adjacent rectangle sides are perpendicular: gradient of M = 2.

1 mark
Find A and the perpendicular gradient

M: y = 2x + c. Substitute A: 12 = 2(–12) + c.

c = 12 + 24 = 36

1 mark
Find the intercept of M

1 mark
Correct equation

y = 2x + 36

(Total for Question 19 is 4 marks)